

KIRAN KUMAR TAMARAPALLI

Data Scientist | ML Engineer

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EDUCATION & CERTIFICATIONS

B.Tech in Computer Science , Lendi Institute of Engineering and Technology CGPA: 7.83/10	2021 – 2025 Vizianagaram, India
Data Science , INNOMATICS RESEARCH LABS	15/04 Hyderabad

TECHNICAL SKILLS

Python Identifiers & Operators, List, Data Types, Data Manipulation, OOPs	Generative AI LLMs, Prompt Engineering, LangChain, LangGraph, RAG (Retrieval Augmented Generation), Vector Databases (ChromaDB), Embeddings, Semantic Search, OpenAI API, Hugging Face
MY SQL and DATABASES MySQL, CRUD Operations, RDBMS, Data Exploration, Data Filtering, DQL, Joins, SQL Clauses, Window Functions, COMMIT & ROLLBACK	ML/DL Supervised & Unsupervised Learning, Neural Networks, Backpropagation, Bias, Variance, Feature Engineering & Extraction, Model Optimization, Algorithms & Statistics
EDA (Exploratory Data Analysis) DataFrames, Series, Data Manipulation, Data Analysis Data Cleaning, Data Wrangling, Handling Missing Values, Filtering, GroupBy Operations Data Merging, Joining, Aggregation, Time Series Analysis, File Handling (CSV, Excel)	TOOLS & PLATFORMS Jupyter Notebook, Google Colab, VS Code, Git, GitHub Streamlit, Hugging Face Spaces, AWS EC2.
LIBRARIES & FRAMEWORKS Pandas, NumPy, Matplotlib, Seaborn, BeautifulSoup, Scikit-Learn	

INTERNSHIPS

AIML Intern , FreshBus • Developing LLM-powered applications using LangChain, embeddings, vector databases, and prompt engineering techniques. • Building Retrieval-Augmented Generation (RAG) pipelines to improve information retrieval and response accuracy. • Evaluating model performance, optimizing prompts, and enhancing AI system reliability for production use cases. • Assisting in the deployment and monitoring of AI solutions to support business operations and customer experience.	04/2026 – Present Hyderabad
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PROJECTS

Real-Time Photo Upload Verification System , Python · PyTorch · timm · ArcFace · OpenCV · NumPy · FastAPI · ONNX • Built an open-set image verification system (MobileNetV3-Large + ArcFace embeddings → per-class prototype + cosine-similarity threshold) that decides whether a user-uploaded photo matches the requested item, achieving 97% accuracy, 0.4% false-accept rate, ~96% junk rejection at ~13 ms/image on 79,000+ real-world phone photos across 22 classes . • Framed the task around its false-accept >> false-reject cost asymmetry and lifted open-set junk rejection from 0% to ~96% via positive-calibrated decision thresholds with a per-class false-accept-rate floor guaranteeing FAR ≤ 5% . • Designed a leak-free evaluation pipeline (pHash deduplication + grouped train/test splits) after diagnosing that duplicate burst photos were inflating reported accuracy, ensuring all metrics were trustworthy. • Benchmarked three model backbones (MobileNetV3, ConvNeXt-V2, DINOv2) under an identical protocol and selected the winner for being 5× faster and 6× smaller at equal accuracy , driving every design choice through seeded, reproducible A/B experiments on a frozen test set.
AI Medical Assistant Chatbot (GenAI Powered) , Python, Streamlit, Google Gemini API, LangChain, Prompt Engineering. • Designed and deployed a production-grade Generative AI-powered medical chatbot providing structured, safety-aware preliminary health guidance with real-time LLM inference via Google Gemini API . • Engineered a multi-step prompt framework categorizing symptom inputs into possible causes, risk severity levels , and recommended next actions — demonstrating responsible AI principles . • Implemented constraint-based prompt logic to prevent unsafe outputs (prescriptions, definitive diagnoses), reducing hallucination risk . • Deployed an interactive application on AWS with secure API key management using environment variables, enabling scalable, production-ready inference .
Used Bike Price Analysis & Market Insights , Python EDA Pandas Matplotlib. Seaborn Statistical Analysis • Analyzed 825+ used bike listings across 10+ Indian states to identify resale price trends, regional market patterns, and key depreciation factors using end-to-end EDA. • Cleaned and preprocessed datasets by resolving 12% missing values and removing duplicate records, ensuring data quality for reliable downstream analysis. • Discovered that single-owner bikes with under 30,000 km usage retained 15–25% higher resale value by correlating ownership type, engine capacity, mileage, and manufacturing year. • Produced 10+ analytical visualizations (histograms, boxplots, scatter plots, heatmaps) to communicate findings effectively to non-technical stakeholders.

CERTIFICATIONS

Python Programming — Innomatics Research Labs	Exploratory Data Analysis Innomatics Research Labs	Power BI (DAX, KPI) — Innomatics Research Labs
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